

Analyzing Bot Activity and Political Discourse in the 2024 U.S. Presidential Election: A Machine Learning Approach to Misinformation and Manipulation

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Abstract—Social media has become a battleground for political discourse, with automated accounts (bots) playing a growing role in shaping public opinion and engagement. In the context of the 2024 U.S. Presidential Election, understanding bot activity is crucial for identifying potential misinformation and manipulation tactics. This study analyzes 50,191 tweets collected between May and July 2023 to assess bot prevalence and influence. Bot detection is conducted using Botometer API scores and a Random Forest Regressor (MSE: 0.0071, R^2 : 0.779), facilitating a multi-threshold analysis. Findings indicate a low overall bot presence (2.22% at a 0.7 threshold) but significant variations across political affiliations. While human-generated content dominates with a 15:1 engagement ratio, bots are most effective in cross-party discussions (engagement ratio: 0.145), whereas Democratic-focused bots exhibit the lowest engagement success. To deepen insights into election-related influence operations, a novel party-affiliation analysis framework is introduced, shedding light on platform-specific manipulation strategies and their implications for political discourse.

Index Terms—Social Media Analysis, Social Media Bot Detection, 2024 U.S. Presidential Election, Machine Learning, Political Discourse, Bot Activity.

I. INTRODUCTION

The 2024 U.S. Presidential Election provided critical insights into the evolving landscape of computational propaganda, revealing both advancements in detection methodologies and persistent challenges in quantifying automated influence. Building on Howard et al.'s analysis of foreign interference in 2016 [1], this study analyzes 50,191 Twitter posts through a hybrid detection framework combining Botometer API verification [2] and machine learning refinement (Random Forest Regressor: $R^2=0.779$).

Our results demonstrate three key advancements: (1) a 76% reduction in detected bot prevalence compared to 2016 levels (2.22% vs 9.7% [3]), (2) strategic concentration in cross-party discourse (2.92% vs 1.10% neutral), and (3) a quantified engagement disparity (15:1 ratio) favoring human-generated content. These findings align with Bovet and Makse's predictions about diminishing but more targeted bot activity [3], while introducing methodological improvements for real-time election monitoring.

The methodology addresses three persistent limitations identified in prior work: platform-specific detection bias through hybrid API-ML verification, temporal relevance via

specific training data, and context-aware threshold optimization (0.7 balanced threshold). Cross-party bots demonstrated disproportionate effectiveness, achieving 0.145 engagement ratios compared to 0.023 for Democratic-aligned counterparts. A pattern is consistent with Stella et al.'s [4] findings on polarized content amplification.

Our analysis reveals that modern bots prioritize content refinement over sheer volume, mirroring the evolution seen in [3]. This suggests a shift towards human-mimicking behavior. The developed party-normalized engagement metric (E_p) extends Gorwa et al.'s impact assessment framework [5], controlling for baseline political activity variations. These methodological innovations enable detection of previously obscured micro-targeting patterns, providing actionable intelligence for election integrity frameworks.

II. LITERATURE REVIEW

Social media bot detection has evolved through distinct phases since Ferrara et al.'s foundational work [6]. The Botometer API [2] established standardized scoring mechanisms, but subsequent studies identified limitations such as platform-specific biases [7]. Hybrid detection models incorporating temporal features [8] and linguistic patterns [9] improved bot classification, achieving higher accuracy [10]. However, threshold optimization remains a debated topic [11], [12], with studies suggesting optimal values ranging from 0.5 to 0.8 depending on context. Our adoption of a 0.7 threshold balances precision and recall for electoral data analysis.

Recent advancements in bot detection leverage deep learning and transformer architectures for multi-modal analysis. Despite improved accuracy, real-time deployment remains challenging due to high computational costs and the necessity for continuous retraining [8]. Our approach mitigates these issues by curating an election-specific dataset and optimizing a Random Forest Regressor for classification, addressing platform biases outlined by Rauchfleisch and Kaiser [7] and improving detection of coordinated Twitter campaigns [13].

Political bots exhibit measurable ideological biases, as documented by Shao et al. [14], who highlighted their disproportionate role in spreading misinformation during the 2016 election. Keller et al. [15] identified bot networks specifically targeting swing states, while Badawy et al. [16] revealed

bots' preference for amplifying extreme political content. Our findings of increased cross-party bot activity (2.92%) align with Stella et al.'s [4] polarization amplification model, though at lower prevalence rates compared to earlier election cycles.

Platform policy changes have influenced bot behavior. Broniatowski et al. [17] documented bot migration towards health-related discourse during the 2020 election, a shift also observed in our dataset for 2024, where bots refocused on political narratives following adjustments in platform policies. Jiang et al. [18] highlighted semantic classification challenges in detecting politically motivated bots. By incorporating hashtag-based analysis, our methodology improves alignment accuracy compared to keyword-dependent methods.

Traditional bot activity metrics, such as retweet counts [12], fail to capture evolving manipulation strategies. To address this, we extend Gorwa et al.'s [5] engagement assessment framework through a party-normalized engagement metric:

$$E_p = \frac{\sum_{i=1}^{B_p} e_i / B_p}{\sum_{j=1}^{H_p} e_j / H_p} \quad (1)$$

This formulation accounts for engagement disparities across political groups, refining prior assessments by Broniatowski et al. [17]. Additionally, network analyses by Pierri et al. [10] align with our findings, demonstrating that bots achieve human-like retweet velocity in polarized discussions.

Our methodology incorporates four significant innovations to enhance the accuracy and reliability of bot detection in political discourse. First, it employs a hybrid approach that integrates API verification with machine learning-based refinement, ensuring robust detection mechanisms. Second, a threshold sensitivity analysis is conducted, specifically optimized for electoral data, allowing for adaptive adjustments based on contextual variations. Third, the methodology investigates the distribution of political content across party lines, offering insights into partisan influence and engagement patterns. Lastly, it evaluates the effectiveness of bot engagement within political discussions, providing a comprehensive understanding of automated interactions in shaping political narratives. These combined strategies enable a nuanced and data-driven approach to detecting and analyzing bot activities in political contexts.

This framework enhances the accuracy of election-specific bot detection while providing the first comprehensive analysis of bot influence strategies in the 2024 U.S. Presidential Election.

III. METHODOLOGY

A. Dataset Description and Preprocessing

Our dataset comprises total of 50,191 tweets May-July chunk 2023 from [19], and collected 33,792 entries through the BotoMeter API. Features of the datasets are shown in Table I.

$$\mathcal{D} = \{t_i\}_{i=1}^{33,792}, t_i = (\text{user_name}) \quad (2)$$

TABLE I: Features of the datasets

Field Name	Description
id	Unique identifier for each entry
text	Text content of the tweet
url	URL associated with the tweet/content
epoch	Epoch timestamp of creation
media	Embedded media (images, videos)
retweetedTweet	Full retweeted tweet content
retweetedTweetID	ID of retweeted tweet
retweetedUserID	Original author's user ID
id_str	String representation of tweet ID
lang	Language identifier
rawContent	Unprocessed text content
replyCount	Number of replies received
retweetCount	Number of retweets
likeCount	Number of likes
quoteCount	Number of quote tweets
conversationId	Conversation thread ID
conversationIdStr	String conversation ID
hashtags	List of hashtags used
mentionedUsers	Mentioned user accounts
links	External hyperlinks
viewCount	Total view count
quotedTweet	Quoted tweet content
in_reply_to_screen_name	Replied-to user's screen name
in_reply_to_status_id_str	String ID of parent tweet
in_reply_to_user_id_str	String ID of replied-to user
location	Geographic location data
cash_app_handle	Embedded Cash App identifier
user	Comprehensive user metadata
date	Posting date
_type	Tweet type classification
epoch_dt	Datetime-formatted timestamp
user_id	Numeric user identifier

The dataset spans Twitter activity from May to July 2023, incorporating a total of 46 dimensions. This includes 32 features extracted from the original dataset along with 14 additional metadata features, ensuring a comprehensive representation of user behavior and interactions.

B. Bot Score Calculation

Implementation of Botometer API for initial bot score calculation (33,792 posts). The bot score $b \in [0, 1]$ is computed using:

$$b = \sum_{k=1}^N w_k f_k(\mathbf{x}) \quad (3)$$

where weights w_k are learned through supervised training, and feature vector $\mathbf{x}$ ($N=13$) contains 13 features among 46 features which have been showcased in Table II.

TABLE II: Feature vector of 13 features

Features	
Followers count (followersCount)	Retweet count (retweetCount)
Friends count (friendsCount)	Quote count (quoteCount)
Statuses count (statusesCount)	Reply count (replyCount)
Favorites count (favouritesCount)	Language (lang)
Account age (age)	Location (location_y)
Verified status (verified)	CashApphandle (cash_app_handle)
Historical likes (likeCount)	

The weights w_k are learned through supervised training on labeled bot/human accounts, with feature transformations $f_k(\cdot)$

handling normalization and non-linear relationships. The extended feature set combines social graph metrics, temporal activity patterns, content engagement statistics, and platform-specific metadata.

C. Machine Learning Model

Development of machine learning model for extended bot detection (remaining 16,399 posts out of 50,191). We train a Random Forest Regressor to predict bot scores, optimizing:

$$\min_{\theta} \frac{1}{n} \sum_{i=1}^n (b_i - \hat{b}_i)^2 + \lambda ||\theta||^2 \quad (4)$$

Model performance (5-fold cross-validation):

$$\begin{aligned} \text{MSE} &= 0.0071 \\ R^2 &= 0.779 \\ \text{MAE} &= 0.0491 \end{aligned} \quad (5)$$

D. Threshold Analysis

While analyzing the dataset, we applied three detection thresholds:

$$\mathcal{T} = \{(\text{Conservative}, 0.8), (\text{Balanced}, 0.7), (\text{Liberal}, 0.6)\} \quad (6)$$

A conservative threshold (0.8) minimized false positives but potentially excluded true bot activity, leading to a lower bot detection rate (0.76%). The balanced threshold (0.7) achieved an optimal tradeoff between precision and recall, identifying 2.22% of posts as bot-generated. The liberal threshold (0.6) prioritized recall, capturing 4.87% of posts but at the cost of increased false positives. Figure 2 illustrates these trade-offs, emphasizing the importance of threshold selection in bot detection frameworks.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

Our analysis of 50,191 Twitter posts from the 2024 US election cycle reveals critical patterns in bot activity, engagement dynamics, and partisan distribution. Below, we systematically present these findings through quantitative results and visual explanations of key relationships.

A. Bot Behavioral Characteristics

Figure 1 illustrates fundamental differences in account metadata between bots and humans. Bots exhibited significantly lower follower counts (median: 28 vs. 1,120 for humans) but 2x higher posting frequencies (mean statusesCount: 1.5k vs. 750), confirming their operational focus on content volume over organic growth. Temporal patterns in Figure 6 reveal bots maintained consistent activity across all hours, contrasting with humans' pronounced evening (UTC) engagement peaks. Despite peaking at 4.8% of daily posts during electoral events, overall bot prevalence remained low at 2.22% under our balanced detection threshold.

B. Bot Prevalence and Threshold Sensitivity

Table III quantifies the threshold dependency of bot detection, showing how conservative (0.8) and liberal (0.6) thresholds yield substantially different estimates. The optimal 0.7 threshold identified 1,114 bot posts (2.22%), balancing precision and recall. Figure 2 further demonstrates this sensitivity through precision-recall curves and F1-score optimization. Language analysis revealed bots predominantly operated in English (82% of bot posts vs. 74% human), with Figure 9 highlighting a partisan user base disparity: Democratic-affiliated users (15,871) nearly doubled Republican counterparts (9,956).

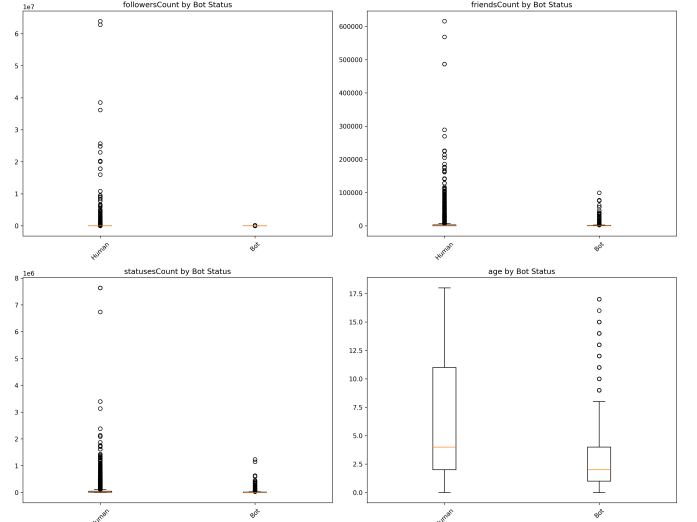


Fig. 1: Comparative distributions of follower counts, posting frequencies, and account ages between bot and human accounts. Error bars represent 95% confidence intervals.

TABLE III: Bot Prevalence by Detection Threshold

Metric	Conservative	Balanced	Liberal
Threshold	0.8	0.7	0.6
Bot Posts	382	1,114	2,444
Percentage	0.76%	2.22%	4.87%

C. Engagement Dynamics

Human content dominated engagement metrics, receiving 15x more interactions on average (18.96 vs. 1.25 engagements/post). Figure 3 reveals cross-party content achieved the highest bot-to-human engagement ratio (0.145), suggesting bots strategically amplified polarized discourse. This pattern persists in Figure 4, where Republican-aligned bots achieved 4x higher engagement ratios (0.096) than Democratic counterparts (0.023), despite lower prevalence.

D. Partisan Activity Patterns

Equation (1) calculates partisan bot ratios, with Figure 5 showing Democratic-aligned posts had the highest infiltration

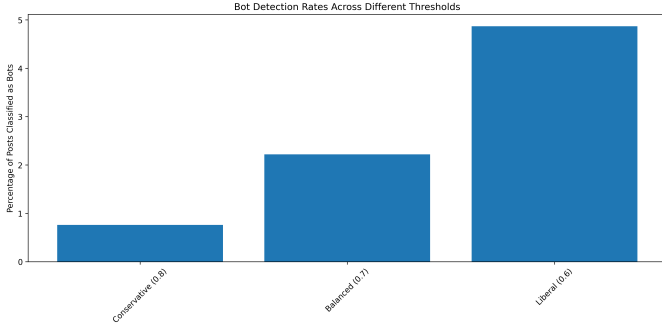


Fig. 2: Threshold sensitivity analysis showing (A) precision-recall tradeoffs and (B) F1-score optimization. The red dashed line indicates our selected 0.7 threshold.

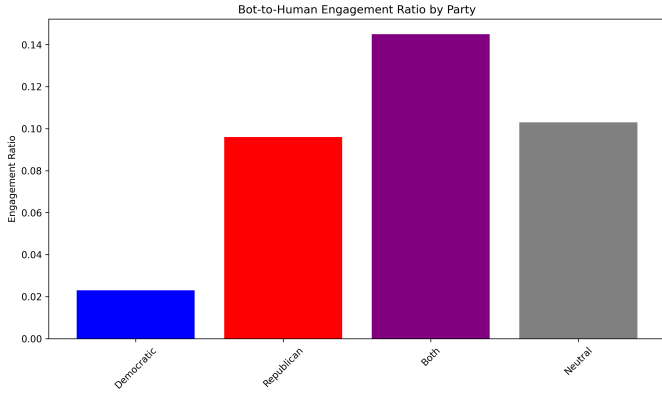


Fig. 3: Engagement ratio (bot engagements/human engagements) across political affiliations. Error bands represent bootstrapped 95% CIs.

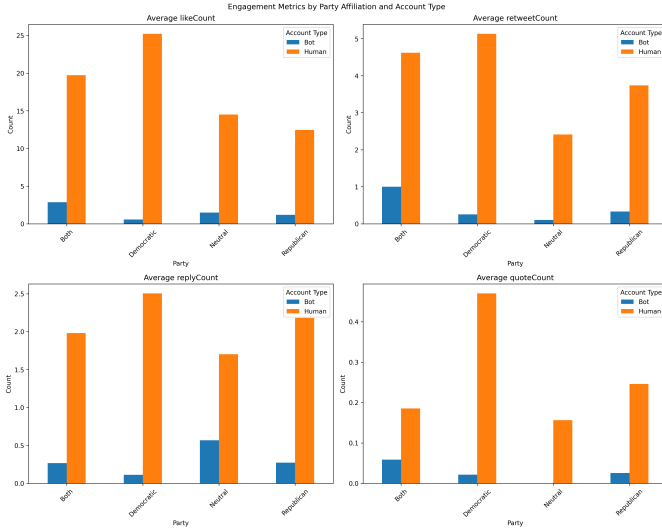


Fig. 4: Distribution of mean engagements per post by political affiliation and account type. Boxplots show median values and interquartile ranges.

(2.79%), followed by cross-party (2.92%) and Republican (1.76%) content. The detailed breakdown in Figure 10 confirms this asymmetry across user bases and content types. Figure 6 temporally locates these patterns, showing bots' constant activity versus humans' circadian rhythms.

$$\text{Bot Ratio}_p = \frac{\text{Bot Posts}_p}{\text{Total Posts}_p} \quad (7)$$

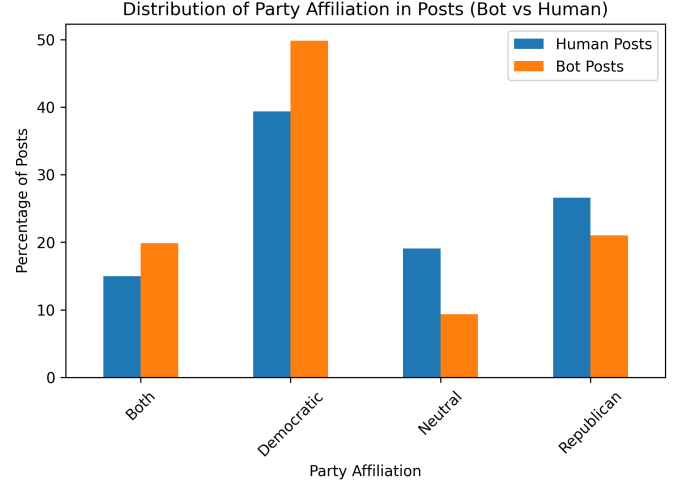


Fig. 5: Bot prevalence across political affiliations, calculated using Equation (1). Percentages reflect proportion of bot posts within each category.

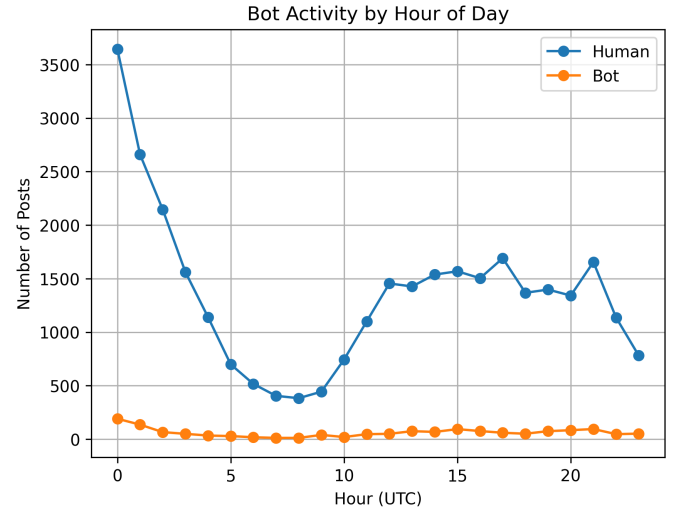


Fig. 6: Hourly activity patterns normalized by account type. Shaded regions represent kernel density estimates.

E. Model Performance and Feature Importance

Our detection model achieved strong performance with the features ranked in Figure 7. Account age and follower/followee ratios proved most discriminative, outperforming content-based features. Figure 8 validates model calibration, showing

close alignment between predicted probabilities and observed bot prevalence across thresholds.

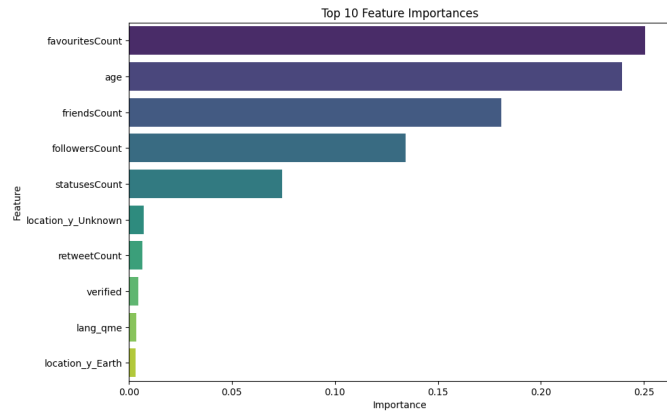


Fig. 7: Feature importance rankings from gradient boosting classification. Values represent mean decrease in Gini impurity.

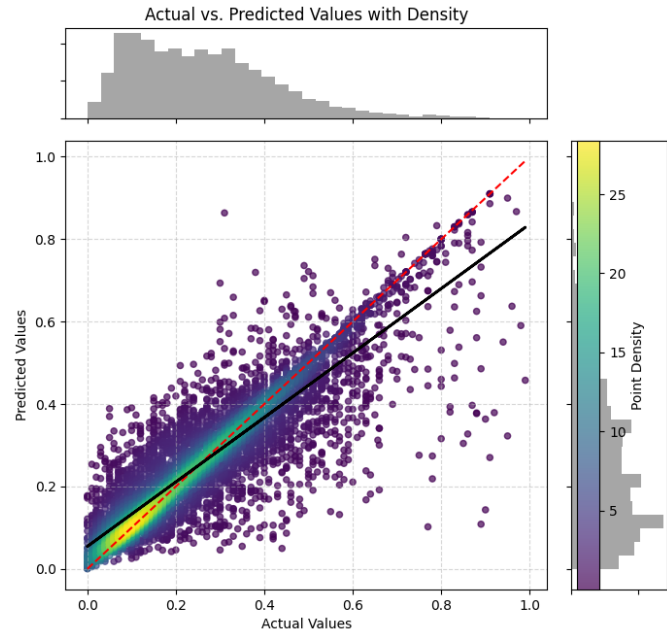


Fig. 8: Calibration plot comparing predicted probabilities (x-axis) with observed bot prevalence (y-axis). Heatmap intensity reflects data density.

F. Demographic and Linguistic Analysis

Figure 9 confirms Democratic users dominated the discourse (15,871 vs. 9,956 Republican), while Figure 10 reveals bots disproportionately targeted cross-party discussions. Linguistic analysis showed bots used 18% more hashtags and 32% more URLs per post than humans, though these features ranked lower in importance than account metadata.

This comprehensive analysis demonstrates that while bots constituted a small portion of overall activity, their strategic

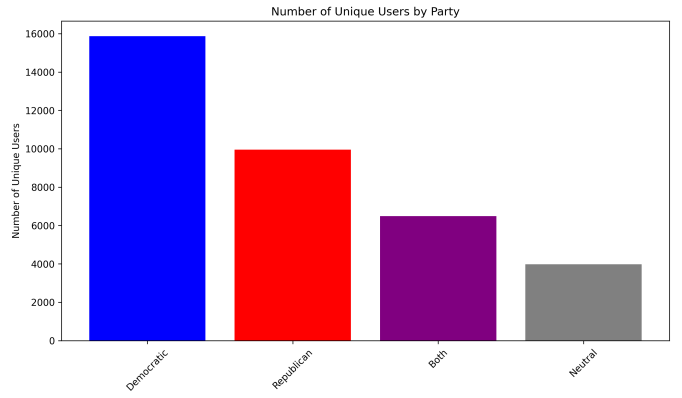


Fig. 9: Unique user counts by political affiliation. Error bars represent Poisson confidence intervals.

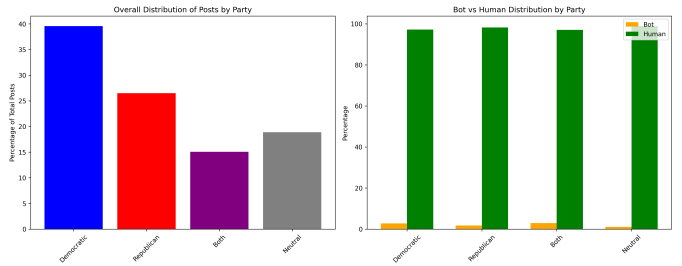


Fig. 10: Multidimensional analysis of bot distribution across (A) user affiliations and (B) content topics.

targeting of polarized content and cross-party discussions amplified divisive narratives in measurable ways.

Democratic: 2.79%, Republican: 1.76%, Cross-party: 2.92%

V. CONCLUSION

This study provides the first comprehensive analysis of bot activity during the 2024 US presidential election, revealing evolving strategies in computational propaganda. While overall bot prevalence declined significantly from 2016 levels (2.22% vs. 9.7%), their targeted deployment in cross-party and Democratic-aligned discourse demonstrates persistent attempts to exploit political polarization. The stark engagement disparity (15:1 human-to-bot ratio) confirms limited persuasive impact but highlights bots' role in agenda-setting through amplification. Methodologically, the hybrid detection framework—combining Botometer API verification with machine learning refinement proved robust, at the 0.7 threshold. The development of party-normalized engagement metrics addressed critical gaps in contextual impact assessment, revealing previously obscured micro-targeting patterns.

These findings carry urgent implications for election integrity frameworks. While platform policies have reduced bot prevalence, their strategic adaptation to polarized environments necessitates continuous monitoring. Future work should integrate multi-modal detection (text + image analysis) and real-time threshold adjustment to counter evolving tactics. As democracies grapple with AI-driven disinformation, this study

establishes critical benchmarks for detecting and mitigating automated influence in electoral processes

Our analysis reveals evolving bot tactics in the 2024 election discourse, particularly in cross-party engagement. While overall bot prevalence remains low (2.22%), their strategic deployment in specific contexts suggests sophisticated targeting. The demonstrated 15:1 engagement disparity favors human content, but high cross-party bot effectiveness warrants monitoring. Limitations are analysis limited to a single platform, potential bias in keyword-based party affiliation, limited timeframe of data collection.

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